

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B.Tech (CE/EE/IT) Examination

December 2010

CL 103 MECHANICS OF SOLID

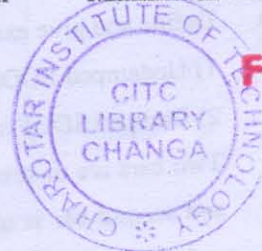
Date: 15/12/2010, Wednesday

Time: 10:30a.m to 1:30 p.m

Maximum Marks: 70

Instructions:

1. All the questions are compulsory.
2. Use different answer books for each section.
3. Figure to the right indicates full marks of the question.
4. Use of calculator is allowed.



FOR REFERENCE

SECTION-I

- Q-1 a Using analytical method determine magnitude and direction of resultant of coplanar concurrent force system shown of Figure 1. (05)
- b State 'Law of Parallelogram of forces'. Derive an expression for the same law with the aid of neat diagram. (02)
- Q-2 a State Lami's theorem and conditions to apply Lami's theorem. (02)
- b ABCDEF is a rectangular hexagon plane. Force 120 N, P, Q, 90 N and 150 N acts along AB, CA, AD, AE and FA respectively as shown in Figure 2. Find force P and Q for the condition of equilibrium of the system. (06)
- c Locate centroid of the area with respect to A shown in Figure 3. All the dimensions are in mm. (06)

OR

- Q-2 a State Varignon's theorem. (02)
- b Find magnitude, direction and location of resultant of the forces shown in Figure 4. (06)
- c Locate centroid of the area with respect to A shown in Figure 5. Refer all the dimensions in mm. (06)
- Q-3 a Explain: a) Coefficient of friction b) Limiting friction force (02)
- b A ladder AB of length 4m and self weight 400 N rest with one end A on horizontal floor and other end B against vertical wall as shown in Figure 6. The co-efficient of friction for floor and wall are 0.4 and 0.25 respectively. Determine inclination of ladder with floor at which it starts sliding. (06)
- c Determine member forces of truss shown in Figure 7 using method of joint. (06)

OR

- Q-3 a Explain assumption made for analysis of pin jointed truss. (02)
- b A block of mass 200 N rest on a rough incline plane as shown in Figure 8. If inclination of plane is 25° with ground and $\mu_s = 0.2$, determine the magnitude and direction of the smallest force P required to start the block up the plane. (06)
- c Find member forces F_{CD} , F_{DG} , F_{GH} of the pin jointed truss shown in Figure 9 using method of section. (06)

SECTION II

- Q-4** a Define: 1) Malleability 2) plasticity 3) Toughness (03)
- b Explain the following with neat sketch of behavior of mild steel bar under tension.
1. Elastic limit and proportional limit (02)
 2. Ultimate load and breaking load (02)

- Q-5** a Define and give example for : (02)
- 1) Undamped SDOF system.
 - 2) Damped SDOF system.
- b Two bars are unstressed and have length 25 cm and 30 cm at 20°C. One bar is of aluminum (06)
- and second bar is of steel as shown in Figure 10. The cross sectional areas of bars are 20 cm² and 10 cm² respectively. Assuming supports are rigid, compute the stress in each member when temperature is 70°C.
- $E_{\text{Aluminium}} = 0.7 \times 10^5 \text{ N/mm}^2$ $E_{\text{Steel}} = 10.1 \times 10^5 \text{ N/mm}^2$
- $\alpha_{\text{Aluminium}} = 10.4 \times 10^{-6}/^\circ\text{C}$ $\alpha_{\text{Steel}} = 12 \times 10^{-6}/^\circ\text{C}$
- c A member ABCD, with three different square cross sections as shown in Figure 11 is (06)
- subjected to concentrated axial force. Calculate load P for equilibrium and compute the net change in length of the member. $E = 200 \text{ GPa}$.

OR

- Q-5** a Define any two - Oscillation, Amplitude, Time period and Frequency. (02)
- b A compound bar ABC is rigidly fixed with wall at two ends as shown in Figure 12. There is (06)
- no stress in the bar at a temperature of 20°C. If temperature rise to 40°C, find the force applied by the rigid wall to the composite bar.
- $A_1 = 200 \text{ mm}^2$, $A_2 = 100 \text{ mm}^2$, $E = 200 \text{ GPa}$ for both and $\alpha = 11.7 \times 10^{-6}$
- c A steel bar ABCD is subjected with axial load as shown in Figure 13. If bar is in (06)
- equilibrium, determine value of force P. Calculate stress in each section and total elongation of bar. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$, diameter of bar is 16 mm.
- Q-6** a Derive an expression for the deformation of tapered bar circular in cross section. (03)
- b Define and explain use of point of contraflexure in beam. (02)
- c Draw S.F and B.M. diagram of simply supported beam shown in Figure 14. (09)

OR

- Q-6** a Derive relationship between shear force and bending moment. (03)
- b Draw S.F and B.M. diagram for cantilever beam of having UDL of intensity 'w' kN/m over (02)
- full span of length 'L'.
- c Draw shear force and bending moment diagram of beam shown in Figure 15. Locate point of (09)
- contraflexion.

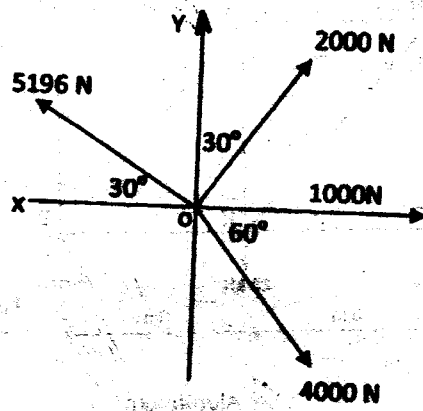


Figure 1

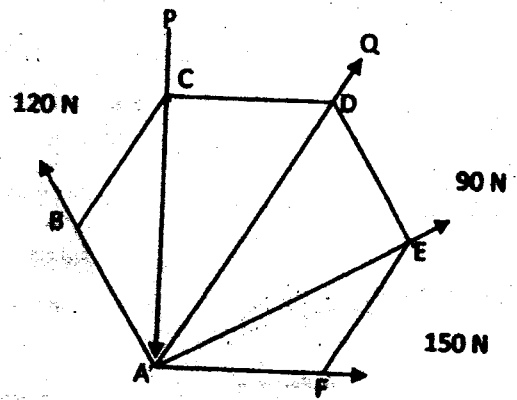


Figure 2

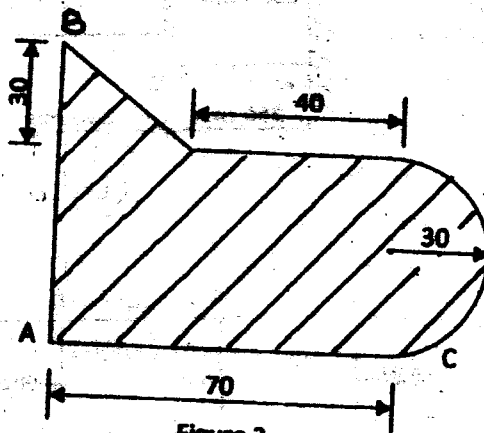


Figure 3

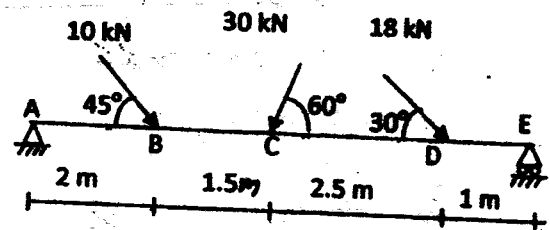


Figure 4

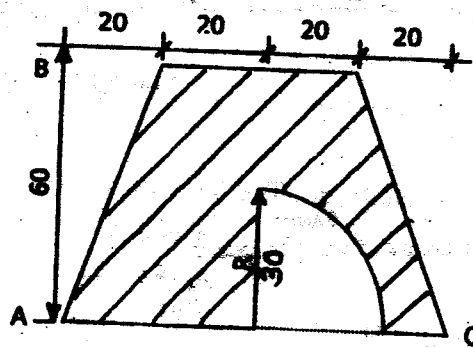


Figure 5

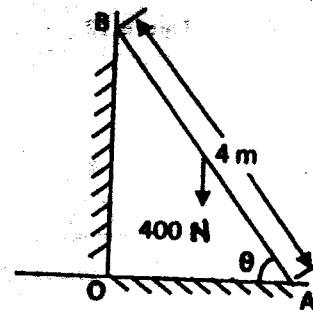


Figure 6

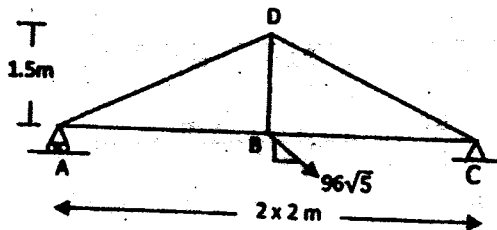


Figure 7

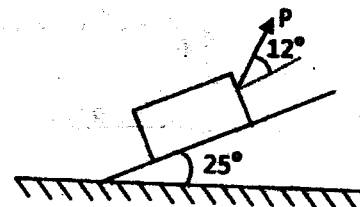


Figure 8

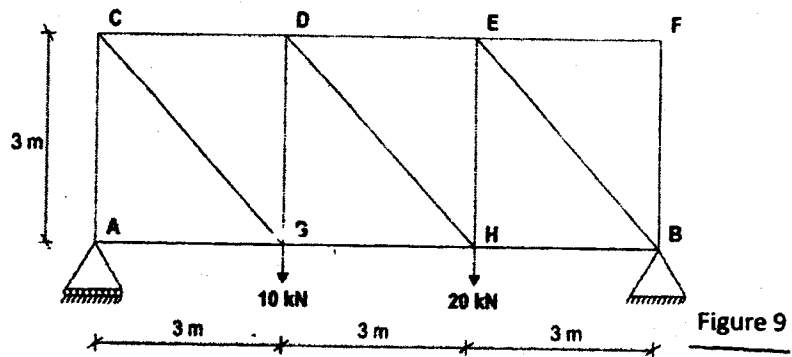


Figure 9

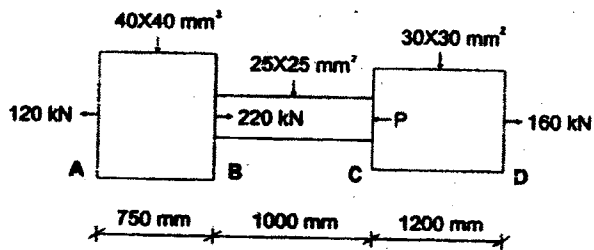


Figure 10

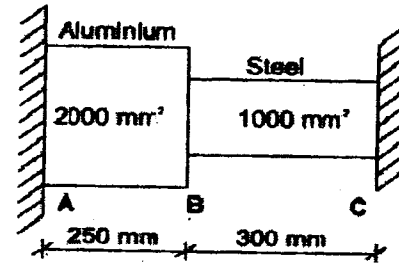


Figure 11

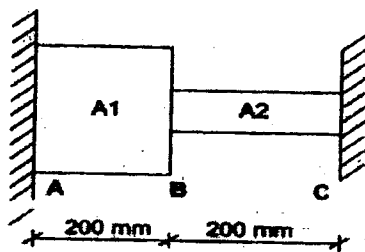


Figure 12

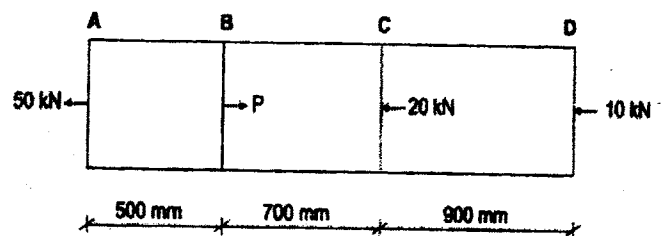


Figure 13

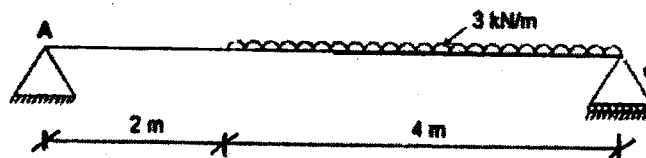


Figure 14

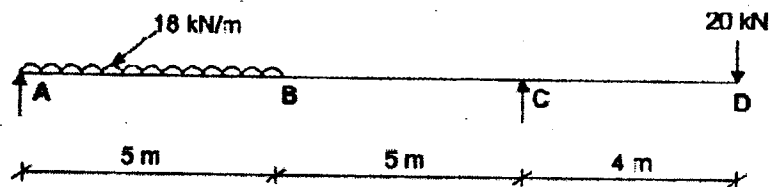


Figure 15